

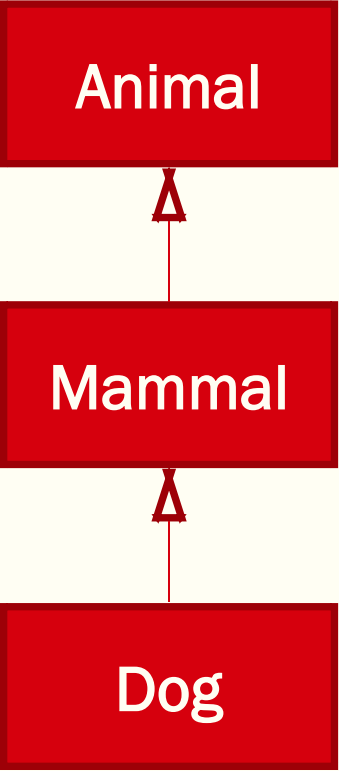


CSC1029 OO Programming

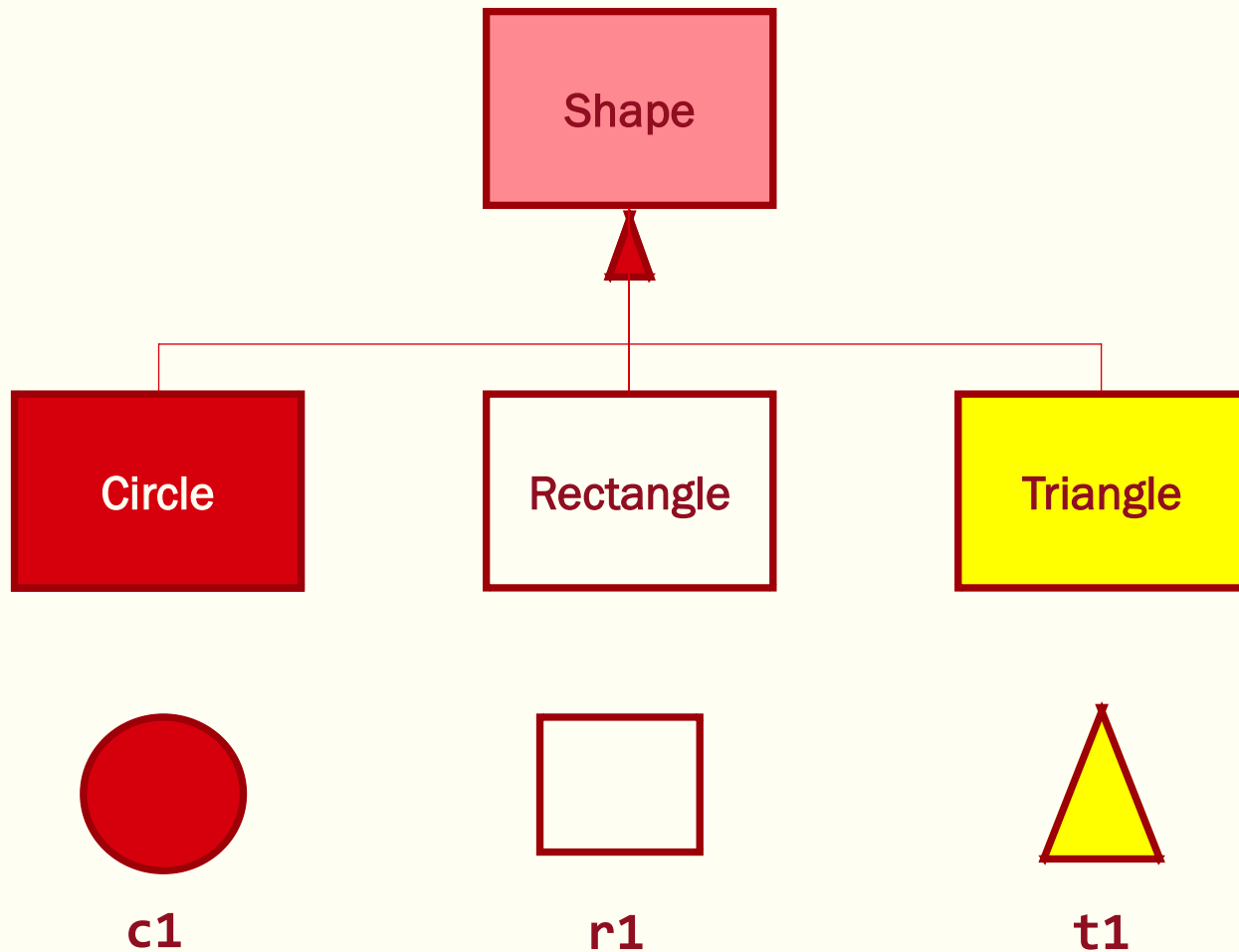
Lecture 20 – Inheritance & Legal Object Referencing



Inheritance and Constructors

An inheritance hierarchy...	Constructors...	What order of execution?
	<pre>public Animal() { System.out.println("Constructing an Animal"); }</pre>	<pre>public static void main(String[] args) { // Order of construction? Dog myDog = new Dog(); }</pre>
	<pre>public Mammal() { System.out.println("Constructing a Mammal"); }</pre>	
	<pre>public Dog() { System.out.println("Constructing a Dog"); }</pre>	Constructing an Animal Constructing a Mammal Constructing a Dog

Inheritance and Object References (1)



```
Circle c1 = new Circle(...);  
Rectangle r1 = new Rectangle(...);  
Triangle t1 = new Triangle(...);
```

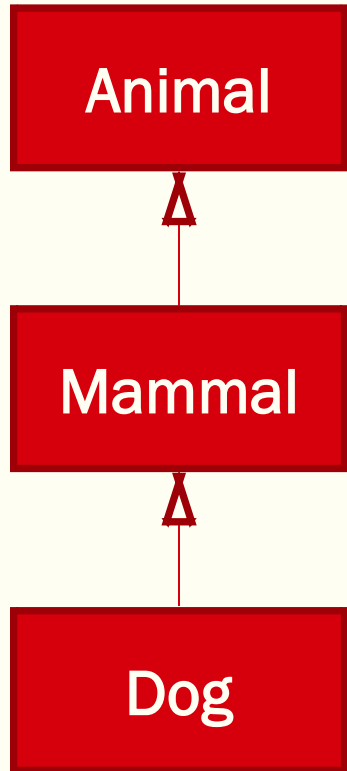
```
Shape s1 = c1;    // safe  
Shape s2 = r1;    // safe  
Shape s3 = t1;    // safe  
s3 = s1;          // safe
```

```
t1 = s2;           // incorrect  
t1 = s3;           // incorrect
```



Inheritance and Object References (2)

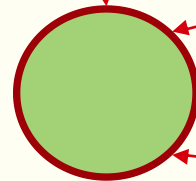
An inheritance hierarchy...



```
Dog myDog = new Dog();
```

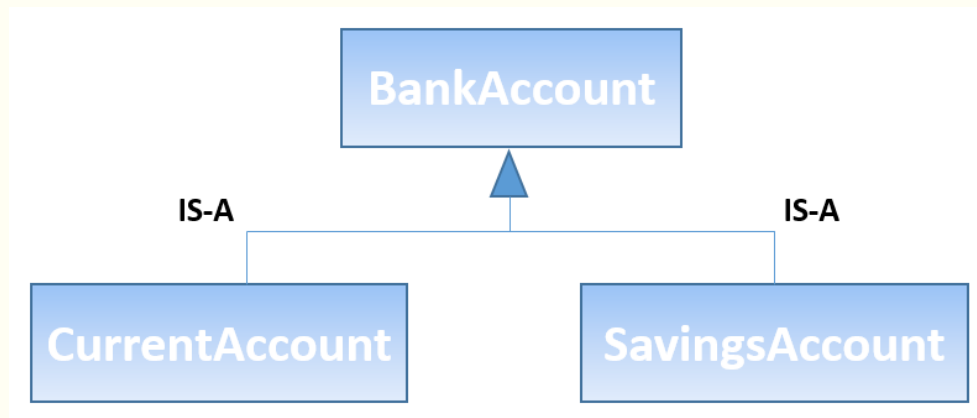
```
Animal aRef1; // a Dog is an  
aRef1 = myDog; // Animal
```

```
Mammal aRef2; // a Dog is a  
aRef2 = myDog; // Mammal
```



```
aRef2 = aRef1; // You can't be sure  
// that an Animal is  
// definitely a Mammal
```

Inheritance and Object References (3)



Consider the following:

```
BankAccount ba1 = new BankAccount( ... );
```

```
BankAccount ba2;
```

```
SavingsAccount sa1 = new SavingsAccount( ... );
```

```
SavingsAccount sa2;
```

```
CurrentAccount ca1 = new CurrentAccount( ... );
```

```
CurrentAccount ca2;
```

Which of the following assignments are legal?

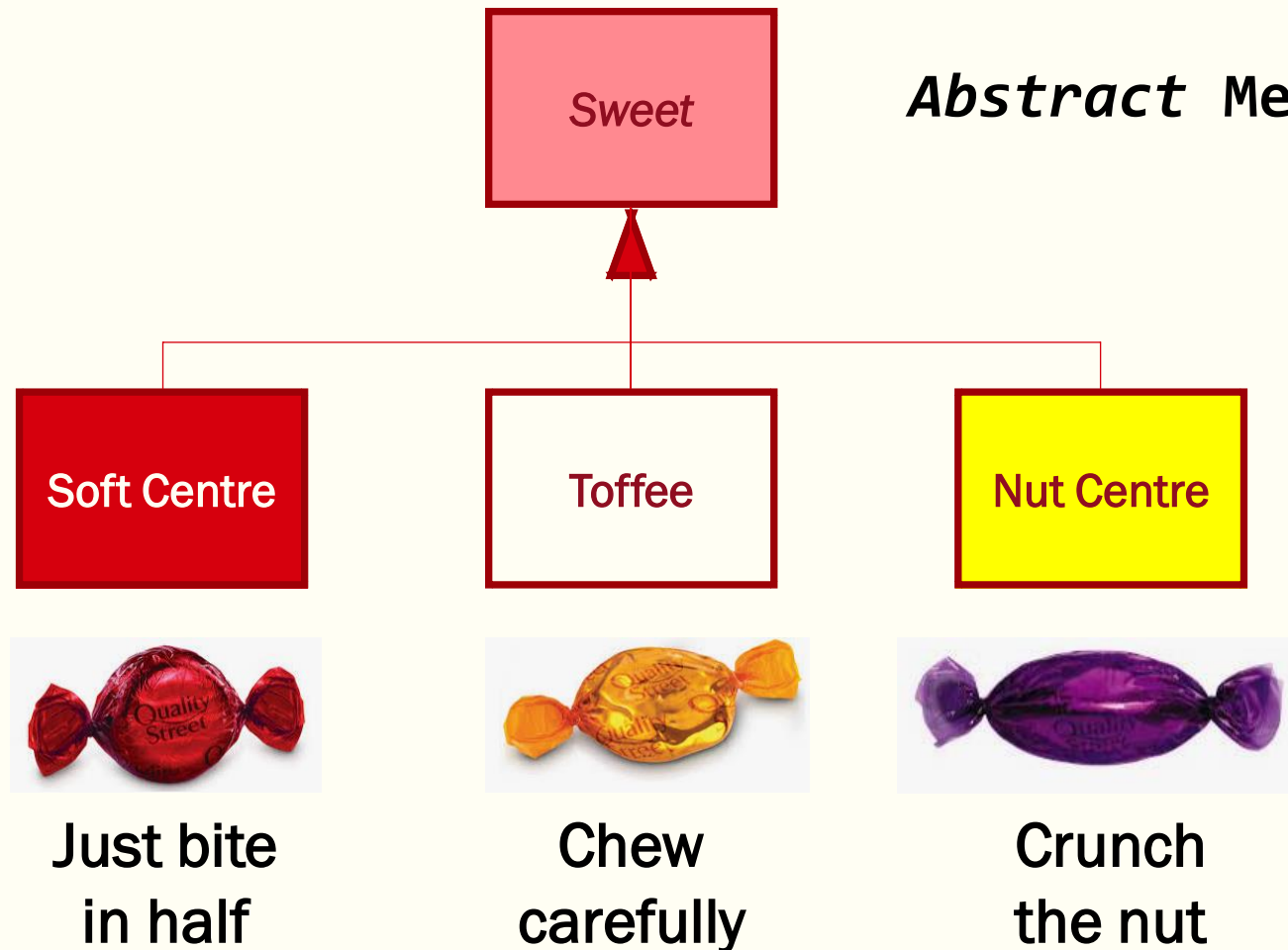
A. `ba2 = ca1;`

B. `ca2 = ba1;`

C. `ca2 = sa1;`

D. `ca1 = ca2;`

Runtime Polymorphism Example



Abstract Method - howToEat



Other sweets are available!



What's Next?

Searching & Sorting